

CompTIA.

# A+ Core 2

Exam 220-1202

A+ 220-1202

# Module 20



## Supporting Mobile Software

## Prepare for A+ Core 2 by:

- Configuring mobile OS security
- Troubleshooting mobile OS and app software
- Troubleshooting mobile OS and app security

# Lesson 20.1

## Mobile OS Security

# Screen Locks (Slide 1 of 2)

Swipe to access

- No authentication

PIN

- 4–6-digit numerical access code
- Password/Passphrase is preferred

Fingerprint

Facial recognition

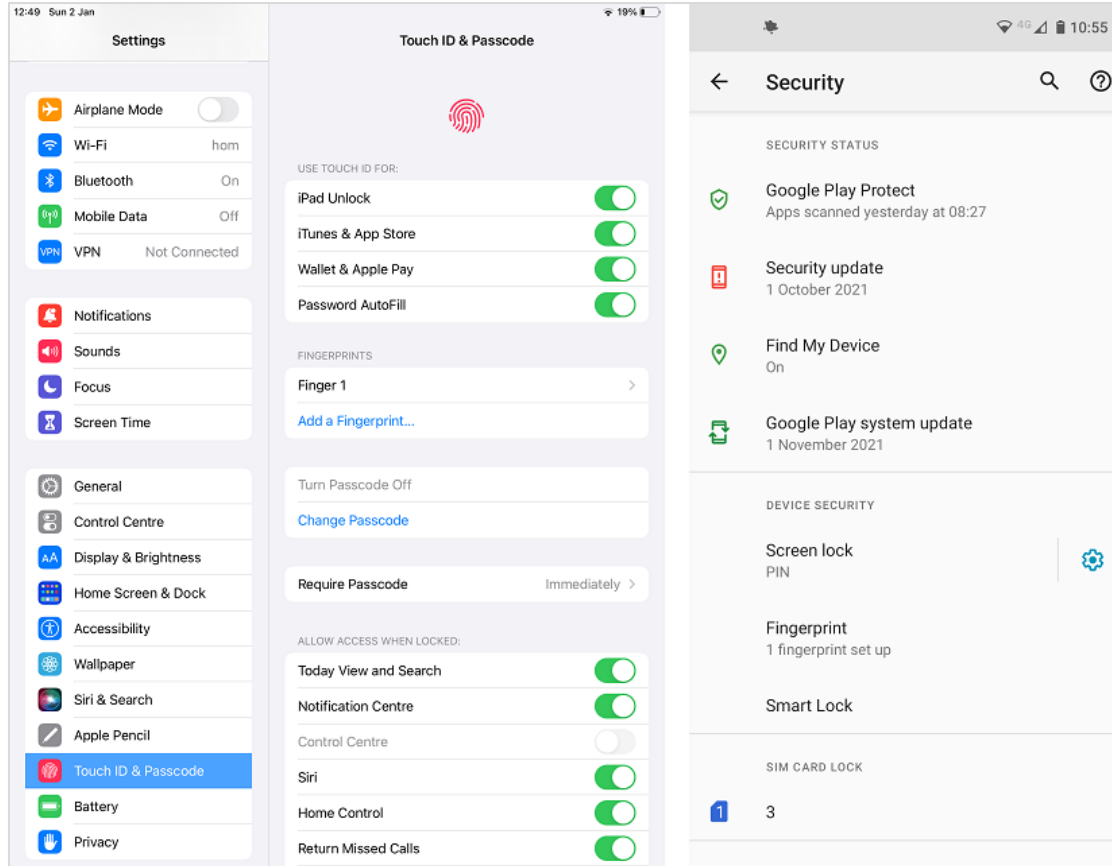
Pattern lock

- Connect the dots on screen in a pattern

Failed login attempts

- Locks device for a period after failed login attempts

# Screen Locks (Slide 2 of 2)



*Screenshots reprinted with permission from Apple Inc. and Android platform, a trademark of Google LLC.*

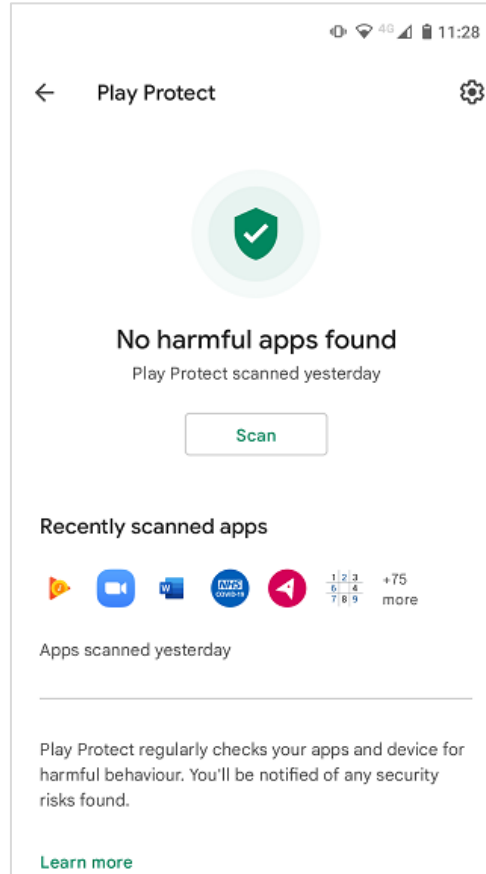
### Patching/OS updates

- Regular updates of OS and applications

### Antivirus/anti-malware

### Firewall

# Mobile Security Software (Slide 2 of 2)



*Screenshot courtesy of Google Play Store, a trademark of Google LLC.*



# Enterprise Mobility Management (Slide 1 of 2)

EMM allows remote management and control of device

BYOD

- Personal device connected to corporate network and systems
- Could require access to the device for monitoring by company

COBO

- Business owned and for business use only

COPE

- Business owned, but personal activities like social media are allowed under AUP

CYOD

- COPE but user can select the device type they prefer (Android or iPhone)

# Enterprise Mobility Management (Slide 2 of 2)

The screenshot displays the Microsoft Azure portal interface for configuring an Android profile. The breadcrumb navigation at the top reads: Microsoft Azure << Device configuration - Profiles > Create profile > Device restrictions > Restricted Apps. The interface is divided into three main panels.

**Create profile**

- Name:** gtslearning EMM Default Android Policy (with a green checkmark)
- Description:** Enter a description... (with a green checkmark)
- Platform:** Android (dropdown menu)
- Profile type:** Device restrictions (dropdown menu)
- Settings:** A blue button labeled 'Configure' is visible.
- Create:** A blue button at the bottom.

**Device restrictions**

Select a category to configure settings.

- General: 11 settings available
- Password: 10 settings available
- Google Play Store: 1 setting available
- Restricted Apps: 2 settings available** (highlighted in blue)
- Browser: 5 settings available
- Allow or Block apps: 3 settings available
- Cloud and Storage: 4 settings available

**Restricted Apps**

Use these settings to stay informed about which users install apps that are not approved for use your company. Select the type of restricted app list:

Prohibited apps - A list of apps that you want to be informed about when users install them. Approved apps - A list of apps that are approved for use in your company. When users install an app that is not in this list, you will be informed.

Type of restricted apps list: Not configured

**Apps list**

Buttons: Import, Export

| App URL                                                        | App Bundle ID    | App Name       | Publisher      |
|----------------------------------------------------------------|------------------|----------------|----------------|
| e.g. https://play.google.com/store/apps/details?id=com.example | e.g. com.example | Not configured | Not configured |
| APP URL                                                        | APP BUNDLE ID    | APP NAME       | PUBLISHER      |

No apps

**OK**

Screenshot courtesy of Microsoft.

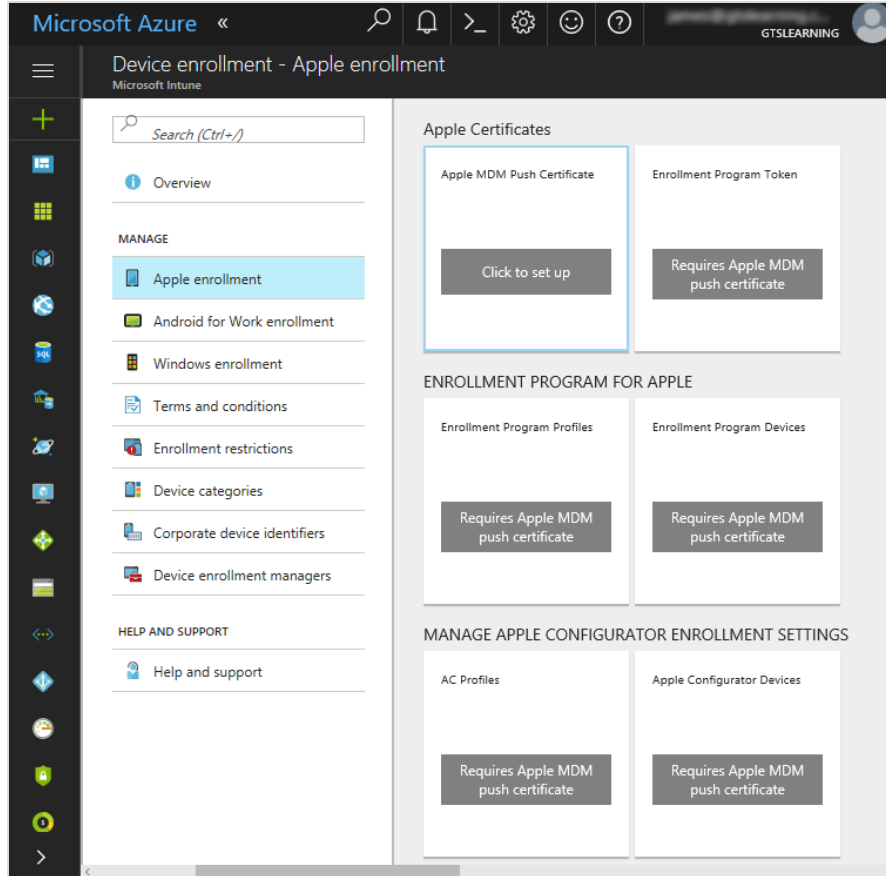
# Mobile Device Management (Slide 1 of 2)

MDM applies security policy to the mobile device

MDM can:

- Manage device settings
- Enforce security policies
- Control app installations and permissions
- Regulate network access
- Monitor compliance
- Facilitate remote management

# Mobile Device Management (Slide 2 of 2)



Screenshot courtesy of Microsoft.

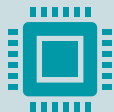
# Two-factor Authentication (Slide 1 of 2)



Two different authentication credentials



Something you are and something you have, etc.



Push notification or  
authenticator application

Microsoft and Google both have their own  
authenticator apps

Many third-party services are available like  
RSA's SecureID

# Two-factor Authentication (slide 2 of 2)

The screenshot displays the Microsoft Azure portal interface for configuring a Work profile on an Android device. The top navigation bar shows the breadcrumb: Microsoft Azure > Create profile > Device restrictions > Work profile settings. The interface is split into three main sections:

- Create profile:** Contains fields for Name (GTSLEARNING EMM Default Android Policy), Description (Enter a description...), Platform (Android for Work), Profile type (Device restrictions), and Settings (Configure). A 'Create' button is at the bottom.
- Device restrictions:** Titled 'Android for Work', it shows 'Select a category to configure settings.' with two options: 'Work profile settings' (12 settings available) and 'Device password' (8 settings available). An 'OK' button is at the bottom.
- Work profile settings:** Titled 'Android for Work', it contains 'General Settings' and 'Work Profile Password' sections. The 'General Settings' section includes: 'Data sharing between work and personal profiles' (Device default), 'Work profile notifications while device locked' (Block/Not configured), and 'Default app permissions' (Device default). The 'Work Profile Password' section includes: 'Require Work Profile Password' (Require/Not configured), 'Minimum password length' (Enter a number (4-16)), 'Maximum minutes of inactivity until work profile locks' (Not configured), 'Number of sign-in failures before wiping the work profile' (Enter a number (4-11)), 'Password expiration (days)' (Enter number of days (1-255)), 'Required password type' (Device default), 'Prevent reuse of previous passwords' (Enter a number (1 - 24)), 'Fingerprint unlock' (Block/Not configured), and 'Smart Lock and other trust agents' (Block/Not configured). A blue 'OK' button is at the bottom.

Screenshot courtesy of Microsoft.

# Mobile Data Security (Slide 1 of 3)

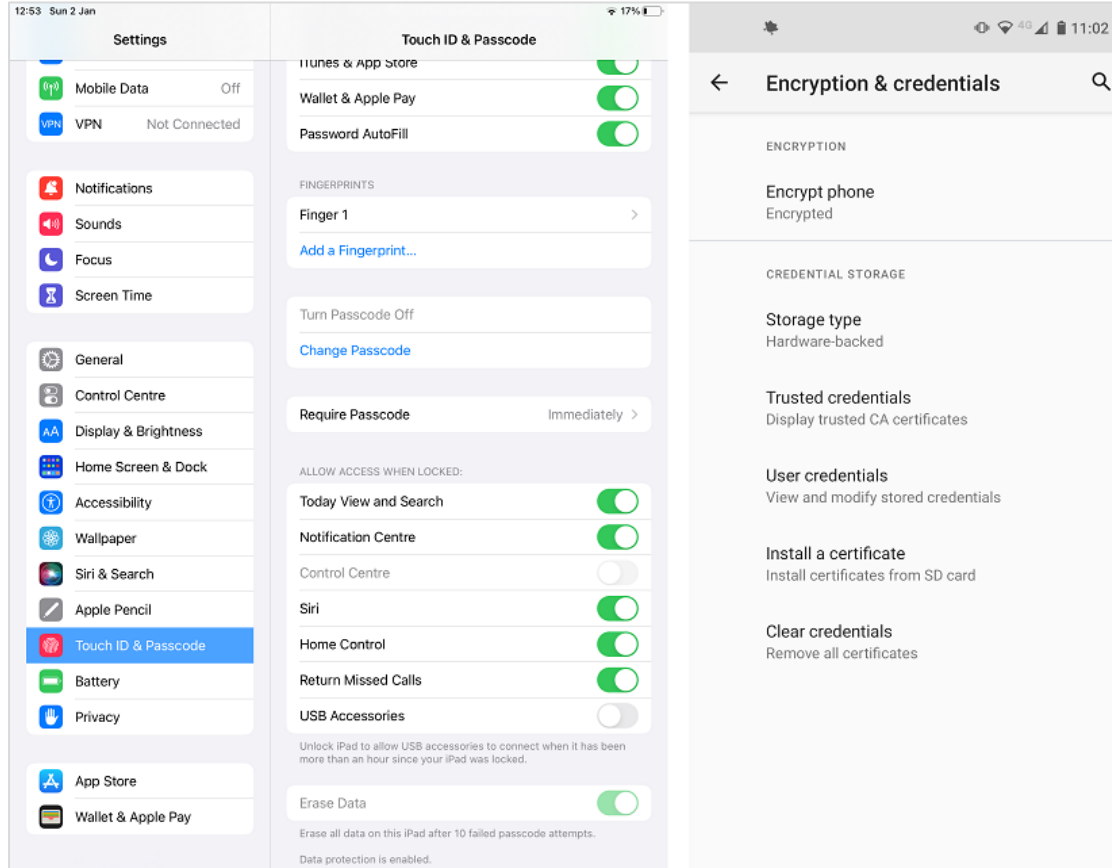
## Device encryption

- Apple- default on and cannot be switched off
- Android- default for file level encryption

## Remote backup applications

- Cloud services
  - iCloud
  - Google Drive
  - Microsoft OneDrive
  - DropBox
- Backup to local computer

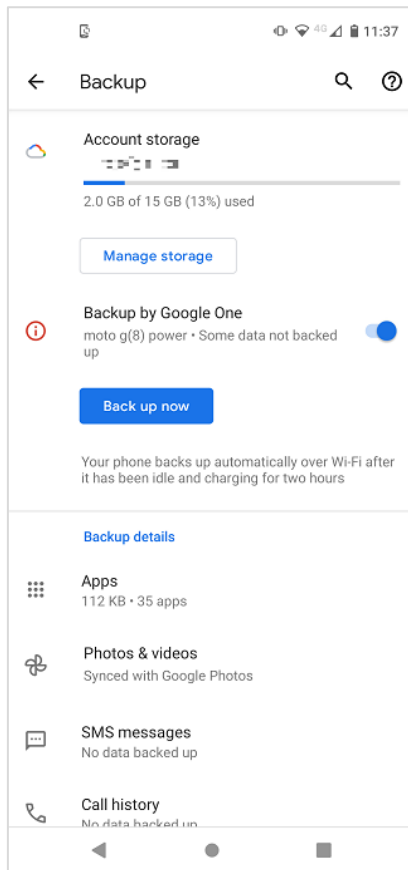
# Mobile Data Security (Slide 2 of 3)



Screenshots reprinted with permission from Apple Inc. and Android platform, a trademark of Google LLC.



# Mobile Data Security (Slide 3 of 3)



Screenshot courtesy of Google One™ subscription service, a trademark of Google LLC

# Locator Apps and Remote Wipe (Slide 1 of 2)



GPS/IPS/Wi-fi/Cellular location services



"Find My Phone/Device" apps for both Apple and Android



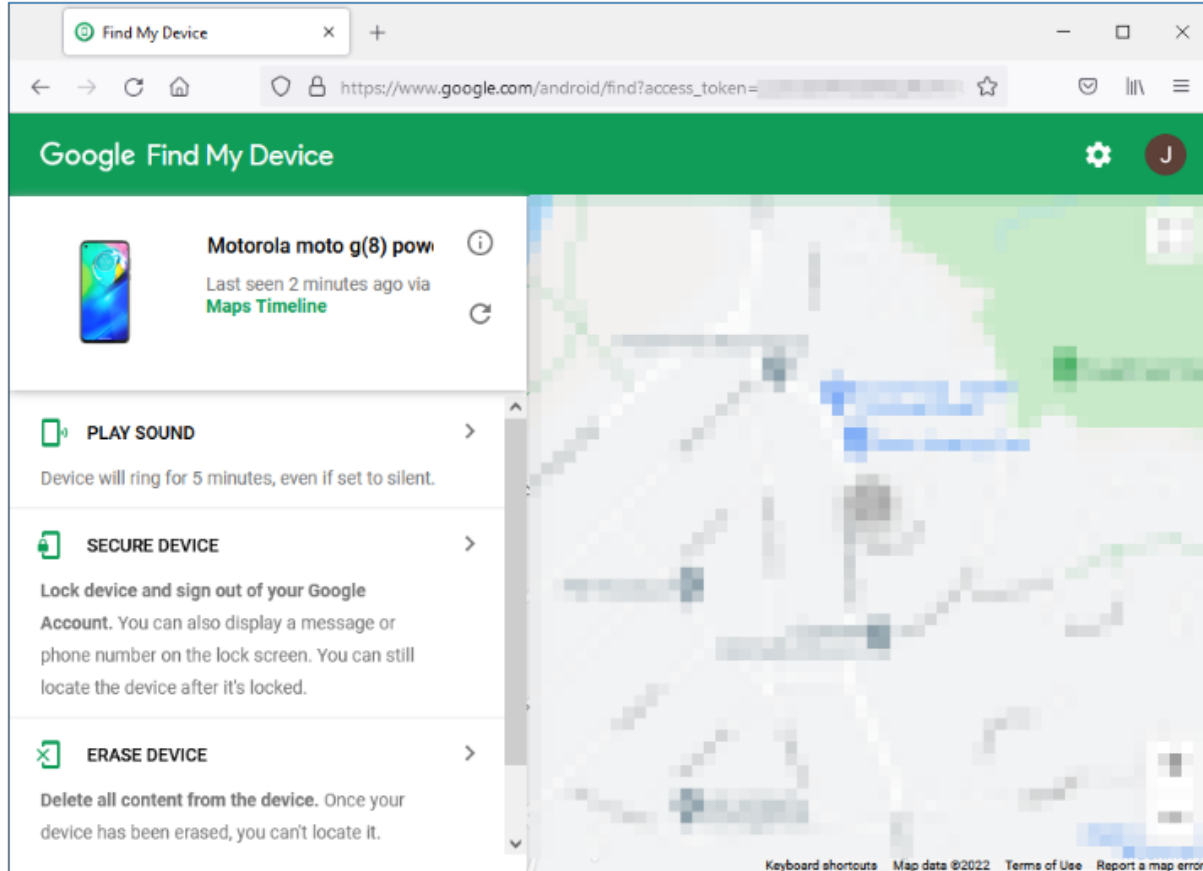
Remotely

Lock

Play sound

Wipe device

# Locator Apps and Remote Wipe (Slide 2 of 2)



Screenshot courtesy of Google, a trademark of Google LLC.

## Lab Activity

- Lab: Configure Remote Wipe  
(available in CertMaster Learn and CertMaster Perform)

## Lesson Review: Mobile OS Security

- Screen Locks
- Mobile Security Software
- Enterprise Mobility Management
- Mobile Device Management
- Two-factor Authentication
- Mobile Data Security
- Locator Apps and Remote Wipe

# Lesson 20.2

## Troubleshoot Mobile OS and App Software

# Mobile Device Troubleshooting Tools (Slide 1 of 2)

- Settings app
  - Down from Top
  - Up from Bottom
- Reboot
- Factory reset
- Battery life
  - Battery health report
  - Smart Charge technology

# Troubleshoot Device and OS Issues (Slide 1 of 2)



OS fails to update



Device randomly reboots



Device slow to respond



Auto-rotate screen not working



# Troubleshoot App Issues



Fail to launch, close, or crashes

Android- Settings > Apps > Force Stop or Disable

Apple- swipe up from bottom of screen, swipe app card up to close



Clear cache

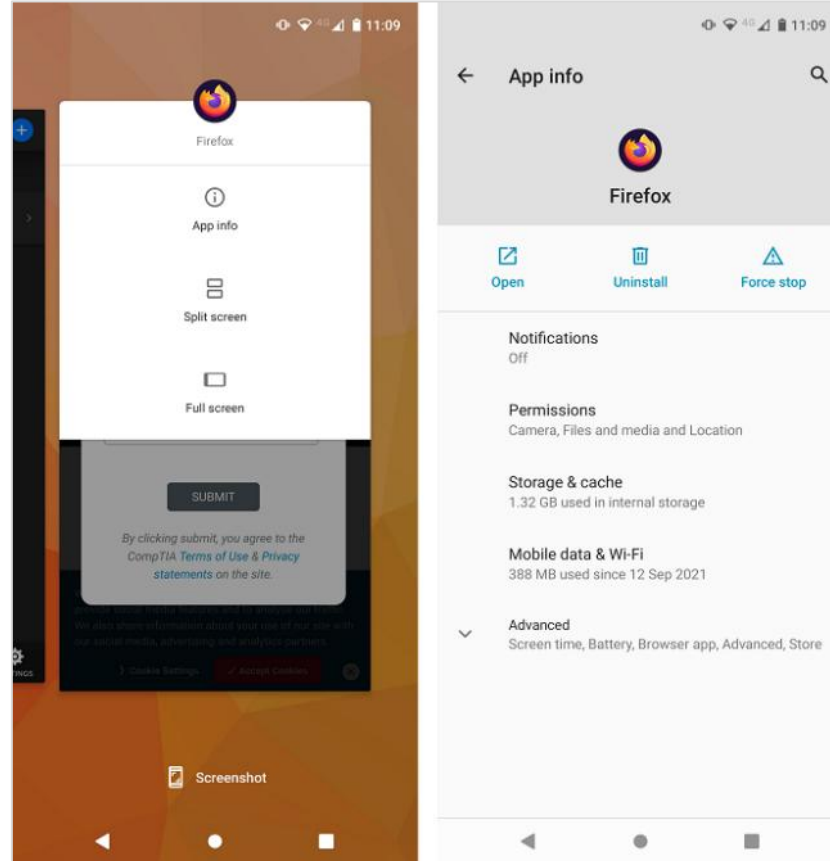


Fail to install

Verify storage and permissions

- Corporate devices may have restrictions
- Confirm connectivity or restriction to download only over wi-fi

# Troubleshoot App Issues



*Screenshot courtesy of Android platform, a trademark of Google LLC.*

# Troubleshoot Connectivity Issues

Signal strength and interference

Configuration issues

Troubleshooting NFC

Troubleshooting AirDrop

- Everyone, Contacts Only, Off
- Verify range to device
  - Bluetooth
- Android has a feature like AirDrop called Nearby Share

## Lab Activity

- Lab: Connect to WiFi  
(available in CertMaster Learn and CertMaster Perform)

## Lesson Review: Mobile OS and App Software

- Mobile Device Troubleshooting Tools
- Troubleshoot Device and OS Issues
- Troubleshoot App Issues
- Troubleshoot Connectivity Issues

# Lesson 20.3

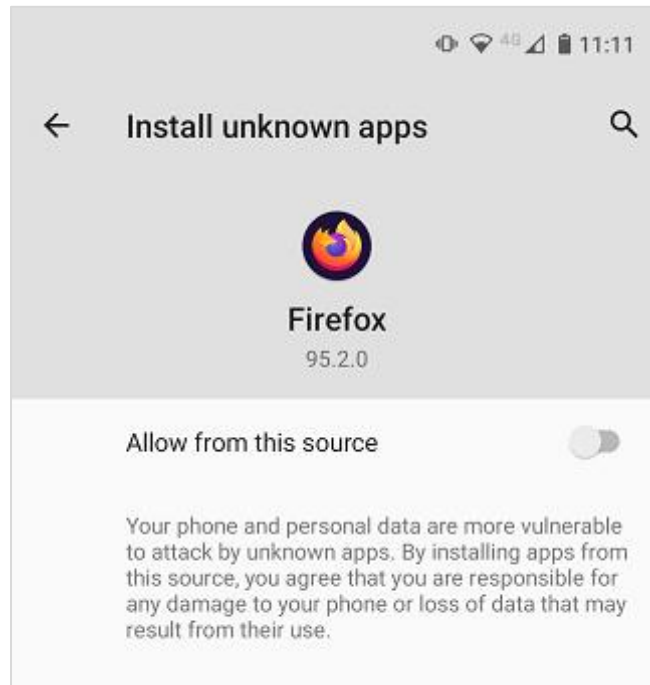
## Troubleshoot Mobile OS and App Security

# Root Access Security Concerns

- Root
  - Primarily refers to Android OS root level access to bypass controls
- Jailbreak
  - Primarily refers to iOS device administrative access to bypass controls
- Developer mode
  - This is not rooting or jailbreaking
  - Specifically used for diagnostic data access and advanced configuration settings

# Mobile App Source Security Concerns

- App spoofing
- Enterprise apps
- APK sideloading
- Bootleg app stores
- Users should only download apps from trusted sources and verify the app digital signature, if possible



*Screenshot courtesy of Android platform, a trademark of Google LLC and Mozilla.*



# Mobile Security Symptoms (Slide 1 of 2)



Excessive ads, fake security warnings



Unexpected app behavior



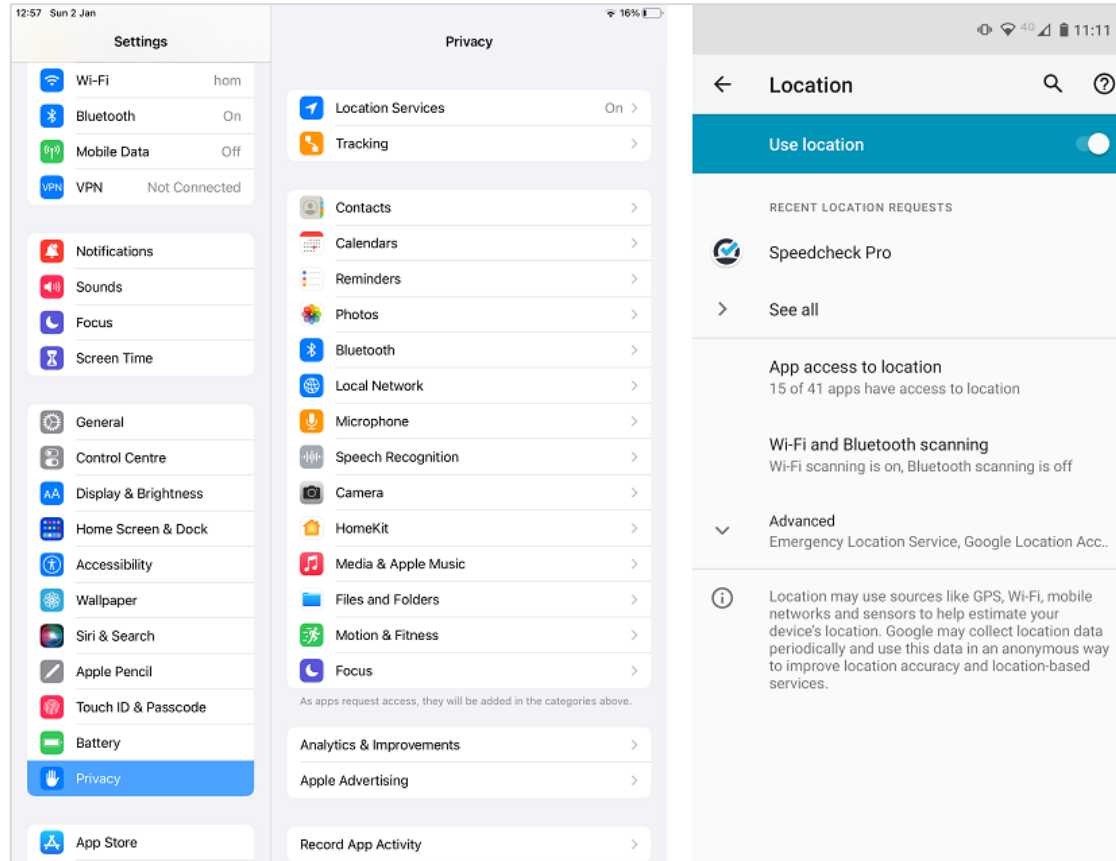
Leaked personal files/data



Location services

Geotagging photos and videos

# Mobile Security Symptoms (slide 2 of 2)



*Screenshots reprinted with permission from Apple Inc., and Android platform, a trademark of Google LLC*

## Lesson Review: Mobile OS and App Security

- Root Access Security Concerns
- Mobile App Source Security Concerns
- Mobile Security Symptoms